Deal or No Deal AI — Multi-Agent Deal Review Pipeline

Every deal has a story. Our agents find the plot holes.

Project 3B demonstrates how specialized agents can support a commercial-loan review from document intake through a source-linked credit memo. This is a local educational application using fictional deals and simulated roles, not an autonomous lending or production approval system.

Problem and solution

Credit reviews require analysts to reconcile loan terms, financial schedules, collateral and policy requirements. Deal or No Deal AI separates those tasks so reviewers can inspect what was extracted, reproduce the calculations, understand exceptions and see what still needs human judgment.

Multi-agent architecture

- **Extraction Agent:** Uses Nebius to extract structured terms and financial inputs with source references.
- **Compliance Agent:** Applies configured educational policies in Python.
- **Risk and Summary Agent role:** Combines deterministic risk flags/available authored ratings with an evidence-linked AI explanation and proposal.
- **LangGraph Orchestrator Agent:** Coordinates state, validated handoffs, conditional stops, persistence and human checkpoints.
- **Supporting services:** Financial calculations and Python memo generation.

New uploaded reviews fan out Financial, Compliance and Risk branches, then join before synthesis. Two normal LLM stages handle extraction and synthesis; other specialist work stays deterministic. Registered samples and historical reviews retain their recorded execution policies.

Product experience

Six connected screens cover Deal Dashboard, New Deal, Review Workspace, Agent Activity, Credit Memo and Decision History. Users inspect uploaded text PDFs, review prefilled reporting context, explicitly run analysis, navigate evidence and record role-controlled comments or decisions. The same Deal ID connects the analysis and report. AI proposals, approval readiness and human decisions remain separate. Educational approval overrides are explicitly labeled and do not clear unresolved requirements.

Demo and evidence

Example Equipment LLC requests a fictional USD1.2 million equipment loan. Its annual DSCR is 1.50x and LTV60%; an illustrative Low score does not resolve an incomplete annual-reporting covenant. Effective readiness remains Pending Review.

An actual uploaded Standard Deal completed live Nebius generation and browser memo download. The expanded general-upload/checkpoint path has offline test coverage; a fresh complete general-profile live walkthrough remains to verify. Recorded acceptance testing passed 57 tests and 55 subtests. These results do not establish broad model accuracy. Live-answer RAGAS remains Phase 2 work; completed saved-answer evaluation is documented separately.

Stack: Python3.12, Streamlit, LangGraph, langchain-core, Nebius Qwen generation, Pydantic, pypdf, Decimal, SQLite and pytest. Runtime grounding uses bounded cited pages, not an asserted production vector-RAG deployment.

Assignment coverage: Three specialist roles, working orchestration, structured agent communication and a sample financial review output.

GitHub: [INSERT LINK — NOT VERIFIED]
Demo recording: [INSERT LINK — NOT VERIFIED]
Application: Local only; no public deployment link verified.